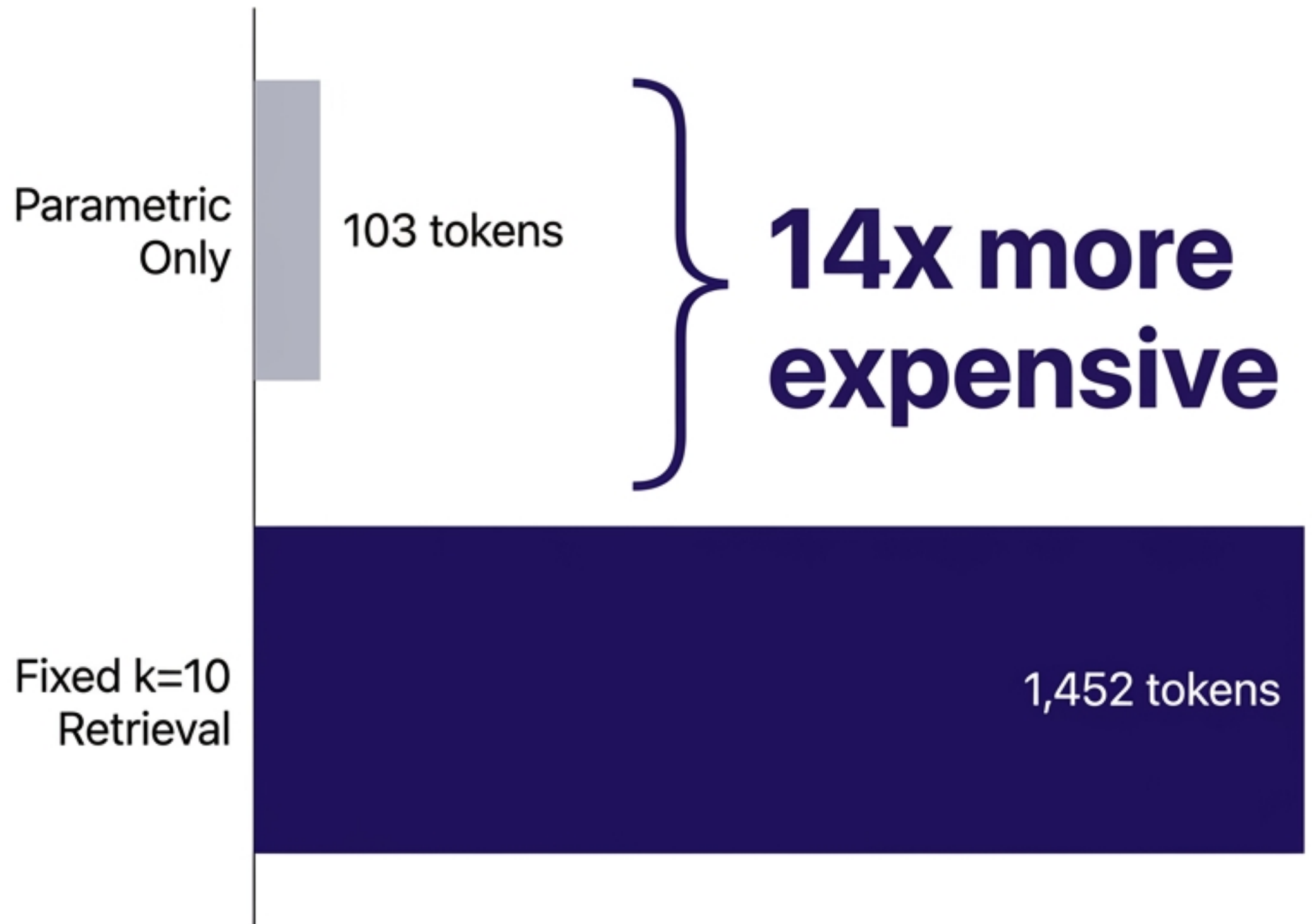


Noise-Gated Retrieval for Cost-Aware RAG

A training-free methodology to cut token costs without killing reasoning quality.

The Heavy Cost of Fixed Retrieval

Most RAG systems blindly retrieve a fixed top-k documents (usually 10) on every query, regardless of how many are actually useful.



At scale, this per-query overhead becomes a severe inference-cost bottleneck.

Why Context Selection is Hard

Distractors are deceptive by design. They are topically adjacent camouflage.

Gold Evidence

In a landmark decision, the committee appointed **Corliss Archer** as the new director, significant **Government Position** for the region. Her background includes notable work with the **Shirley Temple** Foundation.

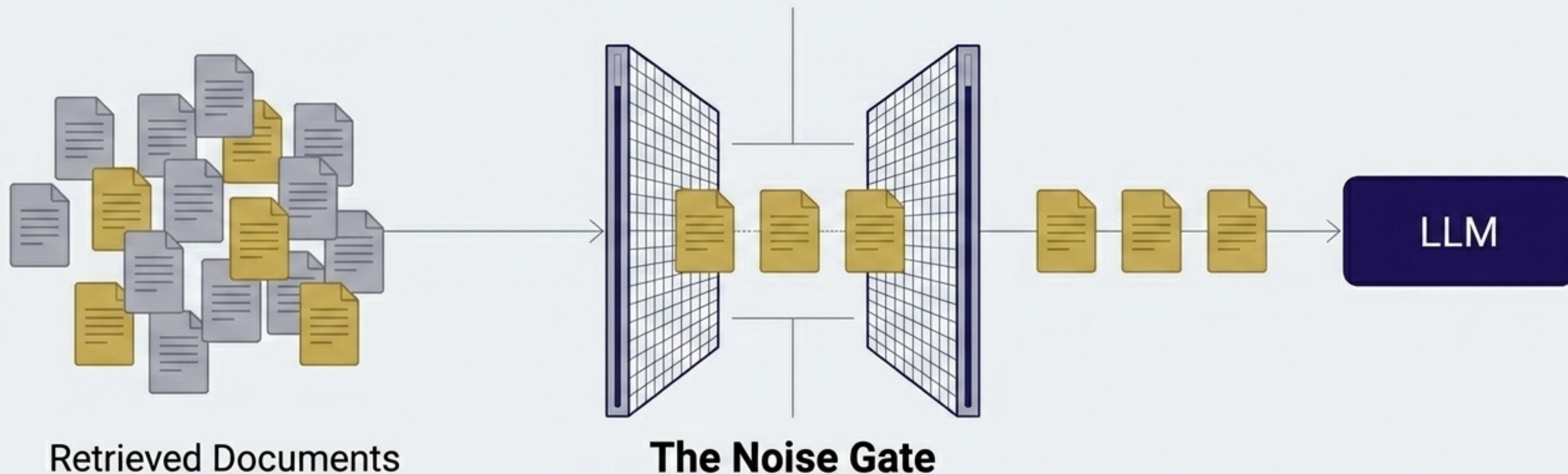
BM25 Distractor

Discussions about cultural icons often reference **Shirley Temple** and her impact. Similarly, the fictional character **Corliss Archer** remains popular. Some argue such roles are akin to a **Government Position**.

Because they share the exact same surface-level vocabulary and entities, standard gap-based truncation or keyword filters fail to distinguish real evidence from “smart” noise.

The Solution: Noise-Gated Retrieval

A lightweight, training-free, two-stage semantic filter

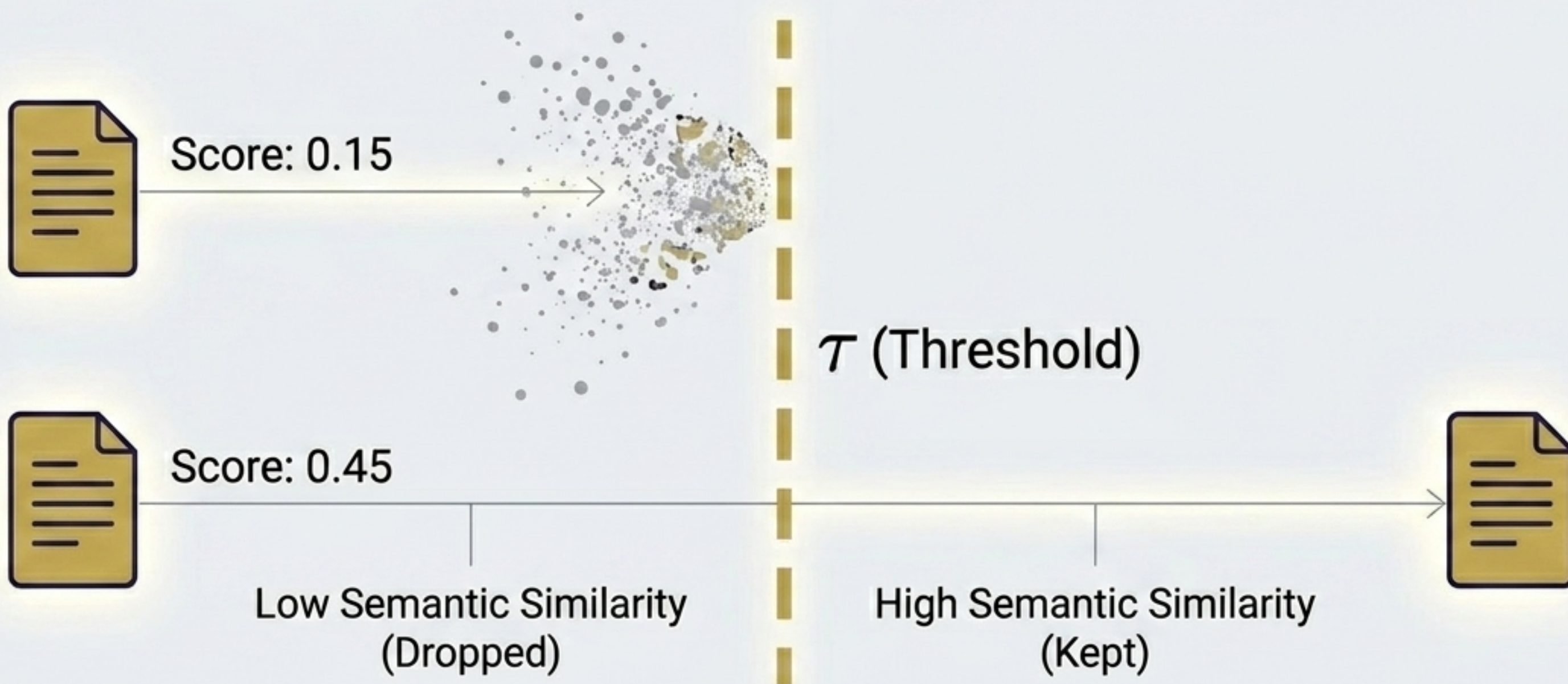


Removes deceptive noise based on deep semantic meaning and redundancy—saving massive token costs without requiring model retraining.

Stage 1: The Relevance Gate

Filter: Cosine Similarity

Does this paragraph actually answer the specific prompt?



Stage 2: The Redundancy Gate

Filter: Jaccard Overlap

Is this paragraph just repeating what we already know?



The Standard to Beat (Fixed $k=10$)

Can we drastically slash the token cost while keeping the quality firmly within a strict 0.05 tolerance?



The Sweet Spot: $\tau = 0.3$

-26.7% Token Cost

Drops from 1,452 to 1,065 tokens per query.



Steady F1 Quality

Score stays strong at 0.764 (well within tolerance)

At a Relevance Gate threshold of $\tau=0.3$, Noise-Gate achieves the “Pareto Knee”—the optimal equilibrium between cost reduction and reasoning preservation.

The Trade-off: When Savings Destroy Quality

Aggressive filtering saves money, but is useless if reasoning breaks down.



$\tau = 0.5$



The Failure of Learned Methods

Complex machine learning methods attempted to beat the baseline, but all failed to match even a basic top-5 retrieval.

Strategy	Tokens Used	F1 Score	Meets Baseline Tolerance?
Noise-Gate ($\tau=0.3$)	1,065	0.764	✓
Adaptive-k	483	0.627	✗
UCB1-TUNED	723	0.704	✗
LinUCB	686	0.620	✗

Diagnosing the Flaw: “Arm Assumptions”



THE MECHANISM:

Bandit algorithms (UCB1) treat document titles as levers to pull, learning over time which titles consistently yield the best answers.

THE FATAL FLAW:

In real-world QA, the questions constantly change. Because **83.8%** of queries encounter titles never seen in training, the model has no historical data and is forced to **blind-guess**.

Diagnosing the Flaw: Lexical Confusability

True Context

Meet Corliss Archer is an American television sitcom and...

Distractor

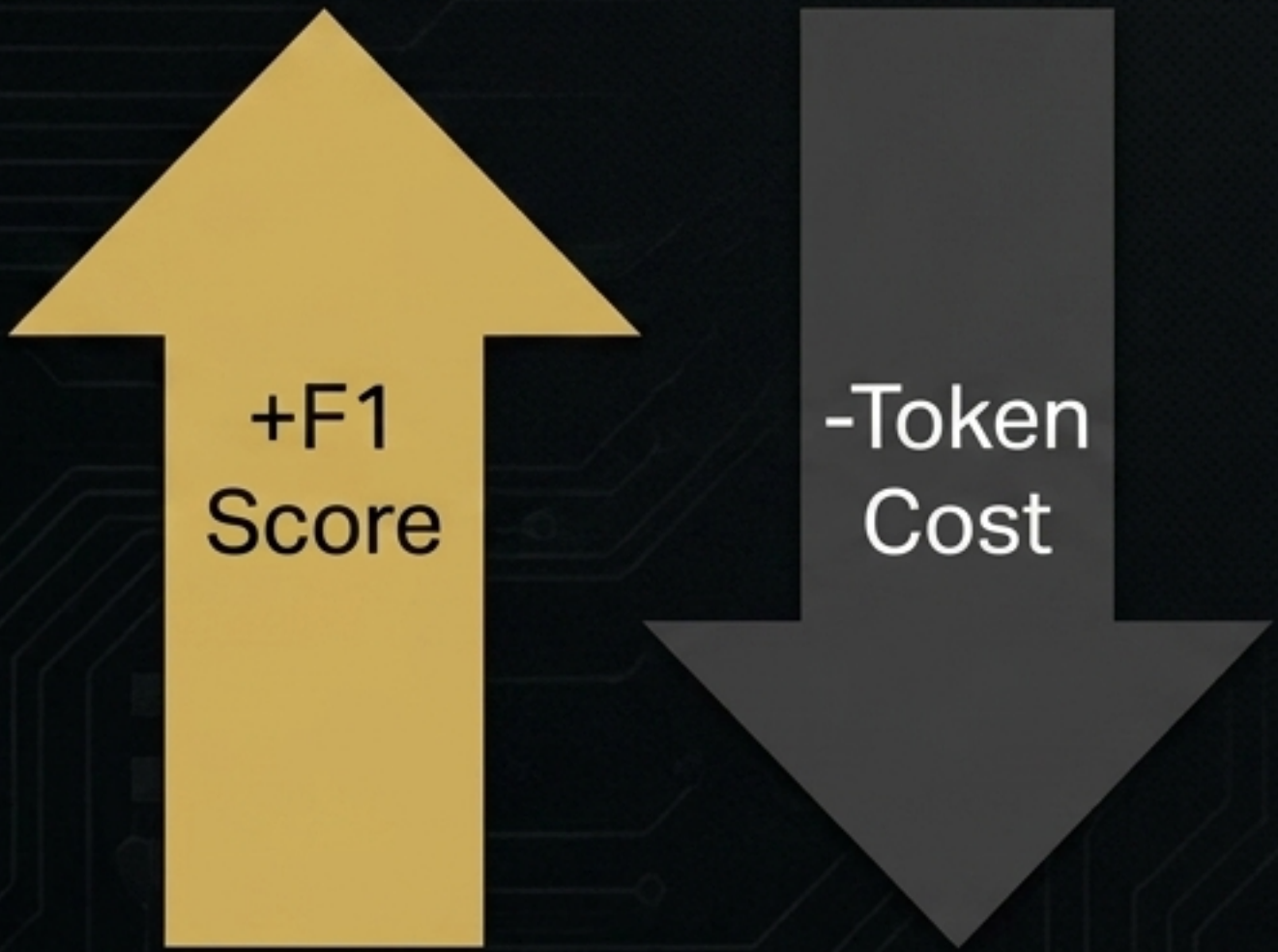
Meet Corliss Archer, a program from radio's Golden Age...

Methods like **LinUCB** and **Adaptive-k** rely on **surface-level vocabulary**. Because 'smart noise' is expertly crafted to share the exact same entities and length as gold evidence, lexical algorithms get confused and cut the actual evidence by mistake.

Generalization Across Domains

Without any retraining, the simple 2-stage filter delivers simultaneous quality gains and cost reductions on entirely unseen datasets.

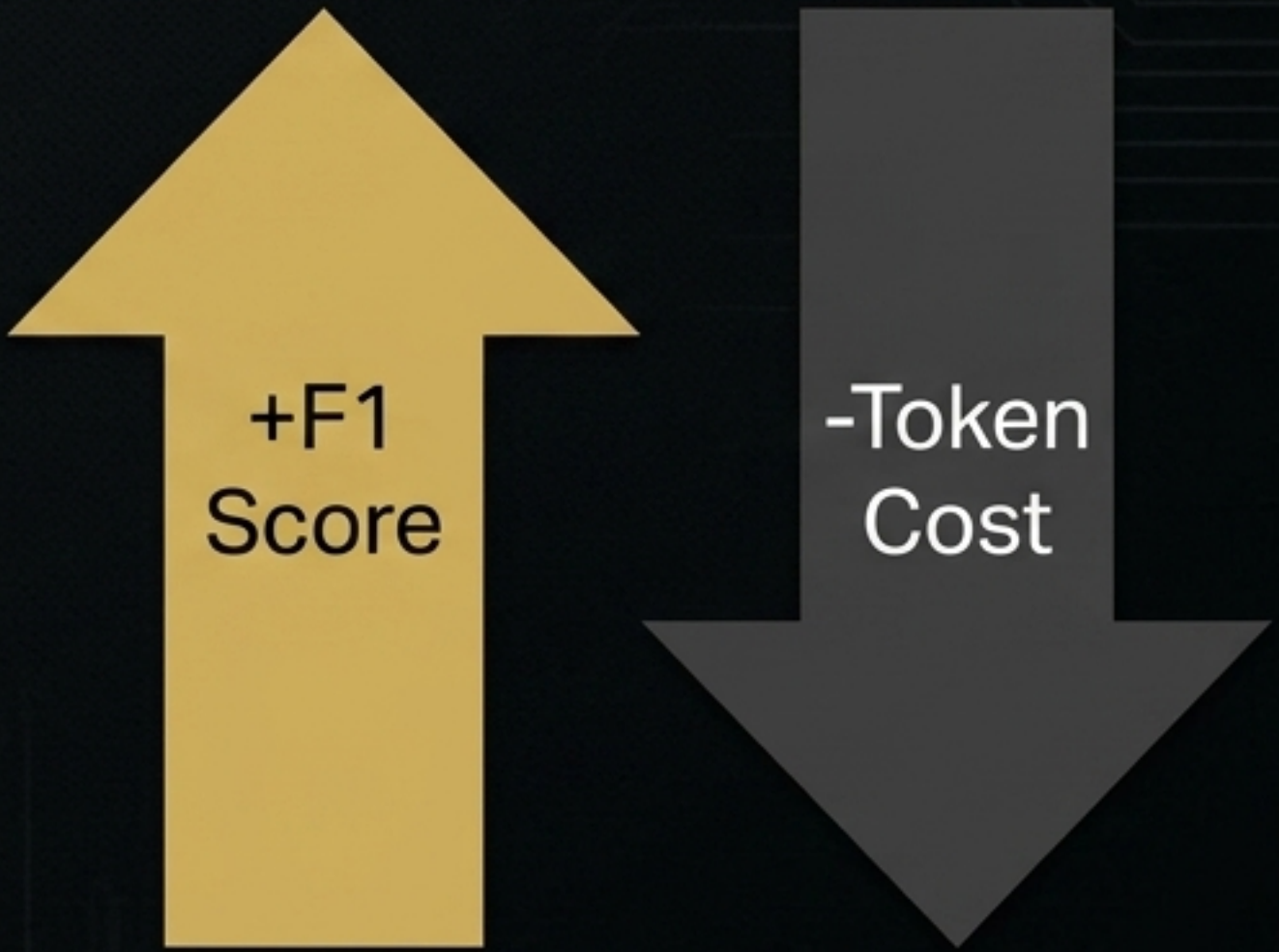
2WikiMultiHopQA



+F1
Score

-Token
Cost

MuSiQue-Ans



+F1
Score

-Token
Cost

The Signal Through The Noise: Key Takeaways



1. The Cost of Status Quo.

Fixed $k=10$ RAG wastes massive amounts of token budget processing deceptive, topically adjacent noise.

2. The Trap of Complexity.

Learned ML models structurally fail at context filtering because dynamic queries break “arm assumptions” and surface-level lexical matching.

3. The Elegant Solution.

Noise-Gate (Cosine Relevance + Jaccard Redundancy) is a training-free, highly effective filter that slashes RAG costs by 26% while preserving reasoning quality.